Software Requirements Specification

Referral Preparation and Follow-through Platform

This draft follows the “Include in the FYP” column in Idea Scoping, with referral-specific medication flagging added to the FYP scope. Detailed workflows and requirements below elaborate that scope. Choices not fixed in the source document are marked as proposed.

1. Purpose

The system shall help a referring clinic prepare a verified referral package, highlight medications relevant to the receiving specialty or planned procedure, share the package with a specialist, resolve missing information, and track the specialist’s response and subsequent follow-up.

It shall combine permitted records from three independently running example systems:

1. A referring clinic system.
2. A laboratory system.
3. A specialist system.

Existing OCR/HTR and language models shall assist with document processing and referral drafting. Clinicians shall remain responsible for verifying and approving clinical content.

2. Project Scope

Included capability	FYP implementation
Three independent example systems	Separate clinic, laboratory, and specialist services, each with its own database and access controls.
Limited common data model	Referral information, relevant medications, laboratory results, and supporting identity and source metadata.
Verified patient linking	Explicitly verified links between institution-specific patient records.
Access controls	Organization, role, and referral-specific permissions, enforced at source systems.
One existing OCR/HTR integration	Extract text from a bounded set of uploaded document types.
Referral brief and evidence	Generate an editable brief with source references and verification status.

Missing-information review	Identify missing, unclear, or conflicting information and track clarification requests.
Referral tracking and return communication	Submission, receipt, specialist response, and follow-up tracking.
Failure handling and audit history	Handle unavailable sources, denied access, processing errors, and recorded user actions.
Referral-specific medication flags	A small set of clinician-reviewed medication-class–specialty/procedure rules, source-linked flags, current-use verification, and specialist acknowledgment.

Proposed prototype boundary: one specialty, English-language records, and a limited set of referral notes, prescriptions, and laboratory reports. These choices require validation with the supervisor and clinical collaborators.

The example systems need only support this workflow; building three complete hospital-management systems is not required.

3. Users and Responsibilities

User	**Responsibilities**
Clinic staff / referral coordinator	Create drafts, upload documents, verify administrative details, coordinate missing-information requests, and track referrals.
Referring clinician	Verify extracted clinical information and current medication use, resolve clinical discrepancies, review medication flags, approve the referral, and review the specialist's response.
Laboratory staff	Maintain example laboratory records and respond to authorized requests for missing or corrected results.
Specialist	Review referrals and medication flags, record flag acknowledgment or request clarification, and approve consultation responses.
Institution administrator	Manage local accounts, permissions, source access, and patient-link corrections.

Administrative access shall not automatically grant unrestricted access to clinical records.

4. End-to-End Workflows

4.1 Create a Referral and Verify Patient Identity

Primary actors: Clinic staff and referring clinician.

1. The user signs in under their institution.
2. The user selects the patient's local record.
3. The system creates a referral draft with a unique referral ID.
4. The user enters the referral reason, destination, specialty, clinician-selected priority, and planned procedure if known; an unknown procedure remains explicitly unknown.
5. The system displays existing verified links to laboratory and specialist records.
6. An authorized user verifies any additional patient links using the agreed identity-verification procedure.
7. The system records the verifier, timestamp, and verification basis.

Outcome: A referral draft linked to the correct local patient and any verified external records.

Exception: If identity is uncertain, external records remain unlinked and shall not enter the referral package.

4.2 Retrieve Records from Participating Systems

Primary actor: Referral platform.

1. The platform requests relevant records through each source's adapter.
2. Each source independently checks the requesting user or service's authorization.
3. Authorized records are returned and mapped to the common model.
4. The platform retains source identifiers, timestamps, and original values.
5. The interface shows the retrieval outcome for each source.

Outcome: An authorized, referral-specific collection of records.

Exception: Unavailable or denied sources are explicitly shown. Their absence shall not be interpreted as an absence of medical history.

4.3 Upload and Verify Paper Records

Primary actors: Clinic staff and referring clinician.

1. The user uploads a supported document and assigns or confirms its type.
2. The system validates the file and attaches it to the selected patient and referral.
3. The existing OCR/HTR integration extracts text.
4. The system proposes structured fields from the extracted text.
5. The user reviews the extracted fields alongside the original document.
6. Unclear fields are corrected, marked unresolved, or excluded.
7. Clinical fields receive clinician verification before being treated as confirmed information.

Outcome: Source-linked, reviewed information available for the referral brief.

Exception: If extraction fails, the user can retry, upload a clearer copy, or manually transcribe information with a source reference.

4.4 Review Missing or Conflicting Information

Primary actors: Clinic staff and referring clinician.

1. The system compares available information with the specialty's predefined referral checklist.
2. It identifies missing items, unresolved extraction, and supported discrepancy types.
3. Users inspect the underlying evidence.
4. Staff create requests for missing information and assign them to the appropriate participant.
5. Responses and additional documents are attached to the request.
6. A clinician resolves clinical discrepancies or explicitly records remaining uncertainty.

Outcome: A checklist showing resolved items and outstanding limitations.

The checklist shall assess referral completeness against agreed requirements; it shall not independently prescribe investigations.

4.5 Referral-specific Medication Flagging

Primary actors: Referring clinician, referral platform, and receiving specialist.

1. Retrieve permitted medication records and extract medication names, doses, dates, and supporting source text from uploaded prescriptions.
2. The referring clinician confirms current use, separating confirmed current medications from historical, stopped, conflicting, or uncertain entries.
3. Map the medication to a verified ingredient and drug class. Leave ambiguous or unmapped names unresolved for review.
4. Evaluate the verified class against a small set of clinician-reviewed specialty/procedure rules. Apply procedure-specific rules only when the required procedure context is known.
5. Display matching flags prominently in the brief with medication, reason, source, and current-use status. Show uncertainty or incomplete-history messages when appropriate.
6. The referring clinician reviews the flag content before approving the package. The specialist records Reviewed or Request clarification against each flag after receipt.

Outcome: Medication documented at an authorized source is highlighted at specialist handover, with traceable evidence and review status. This workflow runs after medication verification and before brief approval; specialist acknowledgment occurs during specialist review.

Illustrative display only; the actual rule requires clinician review:

Medication requires review: anticoagulant recorded

The patient's verified medication list includes an anticoagulant. Review its relevance to the planned treatment.

Source: Clinic A medication record · Current use: confirmed

Actions: Reviewed · Request clarification

If current use is uncertain, display “Possible medication use—please confirm”. If relevant sources are missing, denied, or unavailable, display “Medication history incomplete”. Referral to orthodontics alone shall not imply equal risk for every visit or procedure.

4.6 Generate, Approve, and Submit the Referral

Primary actor: Referring clinician.

1. The system generates a draft brief using selected, permitted evidence.
2. The draft includes source references, a prominent referral-specific medication-flag section, current-use status, and clearly marked unresolved or incomplete information.
3. The clinician reviews and edits the brief.
4. The system checks required administrative fields and outstanding review items.
5. The clinician approves the exact version to be submitted.
6. The system sends the approved package to the specialist system.
7. Delivery and subsequent specialist acknowledgment are recorded separately.

Outcome: A versioned, approved referral package with a traceable submission history.

Exception: A failed send shall retain the approved version and allow a retry without creating duplicate referrals.

4.7 Specialist Review and Return Communication

Primary actor: Specialist.

1. The specialist opens an authorized referral.
2. The specialist reviews the approved brief, medication flags, and accessible supporting evidence, and records Reviewed or Request clarification for each flag.
3. The specialist acknowledges receipt and accepts, declines, or requests clarification.
4. Clarification requests are linked to the referral and answered by the referring clinic.
5. Following consultation, the specialist enters a response.
6. The specialist reviews and approves the response.
7. The system returns it to the referring clinic.

Outcome: A clinician-approved consultation response linked to the original referral.

4.8 Follow-up and Closure

Primary actors: Referring clinician and clinic staff.

1. The clinic receives the specialist’s response.
2. The referring clinician records that it has been reviewed.
3. Staff record follow-up tasks specified by the clinician.
4. Each task receives an owner, status, and an optional due date.
5. Authorized users record completion or a reason for non-completion.
6. The referral is closed after response review and disposition of outstanding tasks.

Outcome: A complete record of referral preparation, specialist communication, and follow-through.

Closing a referral shall represent workflow completion, not a claim that the patient's condition has resolved.

5. Functional Requirements

5.1 Authentication and Authorization

- **FR-01:** The system shall authenticate users and associate each account with an institution and role.
- **FR-02:** It shall enforce permissions on the server for viewing, editing, approving, submitting, and administering records.
- **FR-03:** Each source system shall authorize retrieval requests independently.
- **FR-04:** Access shall be restricted to permitted patients, referrals, and data categories.
- **FR-05:** Disabled accounts and revoked permissions shall prevent subsequent unauthorized access, including through centrally retained data.

5.2 Patient Identity and Linking

- **FR-06:** Each patient record shall be identified using both its institution identifier and local patient identifier.
- **FR-07:** The system shall support verified links between records belonging to the same patient.
- **FR-08:** It shall record the link's status, verifier, verification basis, and timestamp.
- **FR-09:** Identical local IDs or similar names shall not automatically establish a patient match.
- **FR-10:** Authorized users shall be able to invalidate incorrect links, with an audit trail and identification of affected referral drafts or packages.

For example, **Clinic A / MR-102** and **Lab B / MR-102** are separate identifiers until a verified link establishes that they belong to the same person.

5.3 Referral Creation and Management

- **FR-11:** Users shall create, save, reopen, and edit referral drafts.
- **FR-12:** Each referral shall store the patient, referring clinician, source institution, destination, specialty, reason, priority, creation date, and planned procedure when known, otherwise an explicit unknown-procedure state.
- **FR-13:** Authorized users shall search and filter referrals by patient identifier, referral ID, status, destination, and date.
- **FR-14:** The system shall record cancellations and declines with reasons.
- **FR-15:** Changes to approved clinical content shall require a new version and renewed approval.

5.4 Record Retrieval and Common Representation

- **FR-16:** The platform shall retrieve permitted records through defined interfaces for all three example systems.

- **FR-17:** Adapters shall map source records into the limited common model.
- **FR-18:** The system shall preserve original text, values, units, terminology codes where available, and source references.
- **FR-19:** It shall retain clinical event dates separately from upload and retrieval dates.
- **FR-20:** Repeated retrieval of the same source record and version shall not create duplicate clinical entries.
- **FR-21:** Retrieval results shall distinguish successful, empty, partial, denied, unavailable, and invalid responses.

5.5 Document Processing

- **FR-22:** Users shall upload supported PDF and image files within configured limits.
- **FR-23:** The system shall integrate one existing OCR/HTR model or service.
- **FR-24:** It shall retain original documents alongside extraction results.
- **FR-25:** Extracted fields shall begin as unverified.
- **FR-26:** Users shall review and correct extraction while viewing the source.
- **FR-27:** Unsupported or unreadable content shall remain unresolved rather than being silently guessed.
- **FR-28:** Manual corrections shall record the editor and preserve the prior extracted value.

5.6 Medications and Laboratory Results

- **FR-29:** Medication entries shall preserve the documented name, strength, dose, route, frequency, dates, and source where available.
- **FR-30:** The system shall distinguish historical prescriptions from clinician-confirmed current medication use.
- **FR-31:** Laboratory entries shall retain patient identity, test name/code, value, unit, specimen information where relevant, collection time, result status, and source.
- **FR-32:** Missing values or units shall remain explicitly missing.
- **FR-33:** Terminology suggestions shall use a bounded mapping list and require review when uncertain.
- **FR-34:** Unmapped terms shall remain usable in their original form.

5.7 Completeness and Discrepancy Review

- **FR-35:** The system shall apply a predefined checklist for the selected specialty.
- **FR-36:** Checklist items shall indicate whether information is present, missing, unresolved, or clinician-marked not applicable.
- **FR-37:** It shall display potentially conflicting information with both sources and relevant dates.
- **FR-38:** It shall not automatically treat differences across time as errors.
- **FR-39:** Users shall create, assign, answer, and resolve referral-specific clarification requests.
- **FR-40:** Clinical resolutions shall record the reviewing clinician and rationale.

5.8 Referral Brief

- **FR-41:** The system shall generate an editable referral-specific draft using an existing language model.
- **FR-42:** The draft shall cover the referral reason, relevant documented history, medications, results, and outstanding information.
- **FR-43:** Factual statements derived from records shall carry inspectable source references.
- **FR-44:** Newly entered clinical information shall identify its clinician author.
- **FR-45:** The brief shall distinguish verified information, unresolved evidence, and unavailable sources.
- **FR-46:** The system shall not convert “not found” into “not performed” or “no history.”
- **FR-47:** Only an authorized clinician shall approve the referral’s clinical content.

5.9 Submission, Response, and Follow-up

- **FR-48:** The system shall submit only an approved referral version.
- **FR-49:** It shall distinguish delivery, human acknowledgment, and acceptance.
- **FR-50:** Specialists shall be able to request clarification, accept, or decline referrals.
- **FR-51:** Consultation responses shall require specialist approval before return.
- **FR-52:** The clinic shall record receipt and clinician review of the response separately.
- **FR-53:** Users shall track clinician-defined follow-up tasks and record their outcomes.
- **FR-54:** Open requests and overdue tasks shall appear in an in-application work queue.

5.10 Failure Handling and Audit

- **FR-55:** Source failures shall not discard successful results from other sources.
- **FR-56:** Temporary failures shall support bounded retries and visible error states.
- **FR-57:** Repeated submission attempts shall not create duplicate referrals or responses.
- **FR-58:** OCR or language-model failure shall permit manual preparation and review.
- **FR-59:** The system shall audit access, retrieval, identity linking, corrections, approvals, submissions, and status changes.
- **FR-60:** Audit entries shall identify the actor, institution, action, affected record, timestamp, and outcome.

5.11 Referral-specific Medication Flags

- **FR-61:** The system shall use a bounded, versioned set of clinician-reviewed medication-class–specialty/procedure rules. Each active rule shall record its conditions, rationale/reference, reviewer, approval status, and version.
- **FR-62:** It shall preserve the original medication name and source while mapping to verified ingredients and drug classes. Ambiguous, conflicting, or unmapped entries shall require clarification and shall not be presented as confirmed matches.
- **FR-63:** It shall distinguish confirmed current use, historical/stopped use, and uncertain use. A past prescription alone shall not establish current use; historical/stopped medication shall not trigger a current-use flag.

- **FR-64:** The rule engine shall use destination specialty and the planned procedure where available. Missing procedure context shall not be guessed; procedure-dependent rules shall remain unevaluated and visibly require clarification. Specialty-only rules shall provide attention prompts without asserting procedure risk.
- **FR-65:** Matching confirmed current medications shall generate visible referral-specific flags showing medication/class, a plain-language reason, source reference, current-use status, and the rule/version that produced the flag.
- **FR-66:** Uncertain use shall display “Possible medication use—please confirm”. Missing, denied, or unavailable medication records shall display “Medication history incomplete”. These states may coexist with confirmed flags.
- **FR-67:** No match shall be described only as no matching flag within the available verified records and supported rules; it shall not imply that treatment is safe or that no relevant medications exist.
- **FR-68:** Flags shall appear as structured content in the approved referral package and specialist view, independent of AI summary wording. A language model shall not create, suppress, or change a rule’s clinical meaning.
- **FR-69:** Authorized clinicians shall be able to mark each flag Reviewed or Request clarification. The system shall retain the reviewer, timestamp, flag version, and linked clarification request. Reviewed shall mean acknowledgment, not treatment clearance or automatic referral closure.
- **FR-70:** Changes to medication use, mapping, specialty, procedure, or rule version shall trigger reevaluation. Materially changed flags shall require renewed review; approved packages shall remain versioned under FR-15, retaining previous flags and acknowledgments.
- **FR-71:** Medication flags shall not automatically advise stopping, starting, substituting, or changing medication, prescribe treatment, or assign referral urgency.
- **FR-72:** Rule evaluation, failures, source/mapping changes, and acknowledgments shall be auditable. If evaluation fails or coverage is insufficient, the interface shall show the limitation, retain available evidence, and permit manual clinician review without implying a negative result.

6. Referral Status Model

Proposed application statuses:

Status	Meaning
Draft	Referral preparation has started.
Awaiting information	Required information or clarification is outstanding.
Awaiting approval	The package is ready for clinician review.
Approved	A specific version has been approved.
Submitted	The destination system has received the package.

Acknowledged	A specialist-side user has acknowledged receipt.
Accepted	The specialist has agreed to handle the referral.
Response returned	An approved consultation response has been delivered.
Follow-up pending	The clinic has reviewed the response and has outstanding tasks.
Closed	Response review and follow-up disposition are complete.
Declined / Cancelled	The workflow ended with a recorded reason.

Clarification requests shall have their own statuses so that requesting information does not erase the referral's existing progress. Delivery errors shall likewise be tracked separately.

7. Core Data Requirements

Entity	**Minimum information**
Institution and user	Institution ID, user ID, role, account status, permissions.
Patient link	Institution/local ID pair, linked patient reference, verification status and history.
Referral	Patient, sender, destination, reason, specialty, planned procedure or unknown status, priority, status, timestamps.
Source document	Owner, source ID, file reference, type, date, extraction status.
Medication	Original name, verified ingredient and drug class, mapping status/version, dosage details, dates, current-use status and verifier, source record/text reference.
Laboratory result	Test, value, unit, collection time, specimen where relevant, status, source.
Review item	Missing/conflicting information, evidence, owner, resolution, reviewer.
Referral brief version	Content, evidence references, limitations, author/model metadata, approval.

Consultation response	Referral reference, specialist, approved content, receipt/review timestamps.
Follow-up task	Referral reference, description, assignee, due date, status, outcome.
Audit event	Actor, action, target, timestamp, outcome.
Medication rule	Rule ID/version, medication class, specialty/procedure conditions, rationale/reference, clinical reviewer, approval/active status.
Medication flag and review	Referral/brief version, medication/source references, matched rule/version, current-use and evaluation status, flag message, reviewer/time, clarification link.

The central platform shall retain only the referral-related information required by this workflow, with explicit retention rules for summaries, documents, caches, and audit records.

8. Architecture and Reuse Strategy

8.1 Proposed Architecture

Use an established **federated architecture with source adapters**, supported by a modular central application.

- **Clinic service:** Own database, local patient records, referring-clinician information, and source documents.
- **Laboratory service:** Own database, patient identifiers, results, and source authorization.
- **Specialist service:** Own database, incoming referrals, specialist records, and consultation responses.
- **Central referral application:** Coordinates linking, permitted retrieval, document processing, medication mapping, rule-based medication flags, review, approval, and tracking.
- **Background worker:** Executes OCR, brief generation, and retryable retrieval or delivery jobs.

The central application can be a modular monolith to keep development manageable. The three source systems must still run independently and enforce their own access decisions.

8.2 Existing Components to Reuse

Component	**Proposed reuse**
Frontend	React with an existing accessible component library.
Backend	FastAPI with schema validation and generated API documentation.

Databases	PostgreSQL databases separated by source ownership, plus a central workflow database.
Identity	Keycloak for authentication and role integration; application and source APIs enforce record-level permissions.
Data exchange	A deliberately limited FHIR representation with a pinned version.
OCR/HTR	One pretrained model or existing service, selected after testing representative documents.
Brief generation	One existing instruction-following language model with structured, source-referenced output.
Completeness checks	Versioned checklist rules and deterministic validation.
Terminology mapping	A small curated dictionary with reviewed suggestions, including verified medication brand/generic-to-ingredient and drug-class mappings.
Deployment	Containerized services with repeatable configuration.
Medication rule evaluation	A deterministic rule evaluator with versioned clinician-reviewed rules and verified class mappings; reuse existing application validation/configuration components. No prediction-model training.

FastAPI provides OpenAPI-based documentation and validation facilities. Keycloak supports established authentication protocols such as OpenID Connect. FastAPI documentation, Keycloak documentation.

For interoperability, FHIR's ServiceRequest can represent the referral request, while Task supports tracking work toward fulfillment. Application statuses should be explicitly mapped to the chosen FHIR version. FHIR ServiceRequest, FHIR Task.

OCR candidate, not a finalized selection: Microsoft's pretrained trocr-base-handwritten can be evaluated for handwriting recognition. Its model card describes single-text-line input, so full-page processing would require segmentation. Its suitability for your target documents must be measured. Model and download.

8.3 What the Team Must Build

The main custom work shall be:

- Source adapters and common-model mappings.
- Patient-link verification and correction workflows.
- Referral-specific access enforcement.
- Extraction review and source-reference interfaces.

- Specialty checklist, discrepancy-review logic, and clinician-reviewed medication–specialty/procedure rules with flag acknowledgment.
- Brief-generation integration and clinician approval.
- Referral delivery, return communication, and follow-up tracking.
- Failure handling and audit integration.

Reusing models reduces model-development work; it does not remove the need to build and validate these workflows.

9. AI Behavior and Boundaries

Task	Required behavior
OCR/HTR	Propose text from uploaded records; retain uncertainty and originals.
Structured extraction	Populate only supported fields; leave unavailable information missing.
Terminology suggestions	Suggest mappings without overwriting original terms.
Summarization	Draft a referral-specific brief from supplied evidence.
Discrepancy assistance	Flag supported inconsistencies for human review.
Clarification drafting	Suggest questions that users review before sending.

Identity linking, access authorization, medication-rule matching, workflow status, approval, and task completion shall use application logic and authorized user actions. OCR/AI proposes medication details; verified mappings and clinician-reviewed rules determine medication flags. No prediction-model training is required.

Model-generated content shall remain a draft until reviewed. Model outputs shall not autonomously determine diagnosis, treatment, referral priority, or whether a patient needs an investigation.

10. Non-functional Requirements

10.1 Security and Privacy

- Enforce authorization at both the central platform and source APIs.
- Protect network communication using TLS in deployed environments.
- Keep credentials outside source code.
- Restrict document access through authenticated endpoints.
- Define retention and deletion behavior before deployment.

- Use synthetic or appropriately de-identified records for the prototype unless approved data access is available.

10.2 Reliability and Integrity

- Preserve drafts when processing or retrieval fails.
- Prevent silent overwriting of approved versions.
- Detect concurrent edits before saving conflicting changes.
- Preserve event order and timestamps.
- Demonstrate backup and restoration of workflow data.
- Ensure one unavailable source does not make the entire application unusable.

10.3 Usability

- Display original evidence alongside extracted information.
- Make missing, unverified, and conflicting information visually distinct.
- Provide clear next actions and understandable error messages.
- Support keyboard navigation and avoid relying on color alone.

10.4 Performance

Proposed test targets, not established results:

- Ordinary screen loads and saves: within 2 seconds for at least 95% of requests under a documented 10-user test load.
- Source retrieval: configurable timeout, initially 10 seconds per source, followed by a visible outcome.
- OCR and brief generation: asynchronous processing with visible job status; completion targets set after hardware and document-size benchmarking.

10.5 Maintainability

- Separate model integrations from workflow logic.
- Version API schemas, checklist rules, medication mappings and flagging rules, prompts, and model configurations.
- Allow the chosen OCR or language-model component to be replaced through a defined interface.
- Document supported document types, terminology coverage, and known limitations.

10.6 Medication-flag Reliability and Usability

- The same verified inputs and rule version shall produce the same flag results; identical medication/rule matches shall not create duplicate visible flags.
- Flags shall be prominent, understandable without color alone, and linked to permitted supporting evidence. Uncertain use, incomplete history, and confirmed matches shall remain distinguishable.

- Only clinician-reviewed rules and verified mappings shall be activated. Administrative access alone shall not confer clinical rule approval; rule configuration may be maintained as versioned files for the FYP.
- The prototype shall document supported medication classes, specialties, and procedures. A small curated rule set is sufficient; comprehensive drug-interaction checking and prediction-model training are not part of this feature.

11. Acceptance Criteria

The FYP shall be considered functionally complete when the following can be demonstrated:

Area	Acceptance test
Independent systems	Clinic, laboratory, and specialist services run with separate databases and source-level authorization.
Identity	Two patients with identical local IDs in different institutions remain separate unless explicitly verified as the same person.
Authorization	Unauthorized requests are denied at the source and central interfaces.
OCR review	A supported document is uploaded, extracted, corrected, and clinician-verified with its source retained.
Data integrity	Medication details and laboratory values retain original values, units, dates, and provenance.
Missing information	An incomplete referral creates a review item that can be assigned, answered, and resolved.
Conflicting information	Both pieces of evidence remain visible, with a recorded clinician resolution or unresolved status.
Brief approval	An unapproved brief cannot be submitted; changing approved content requires renewed approval.
Follow-through	A referral is received, acknowledged, responded to, reviewed by the clinic, and closed after task disposition.
Failure recovery	A source outage or model failure leaves usable results and supports recovery or manual completion.
Duplicate prevention	Retrying a submission does not create a second referral.

Audit	The demonstration's identity, access, correction, approval, and communication actions are traceable.

11.1 Medication-flag Acceptance Tests

- A confirmed current medication, verified class mapping, and matching approved specialty/procedure rule produce the expected flag in the referring clinician's draft and the specialist's approved package, with source and rule version.
- A historical or stopped prescription does not produce a confirmed current-use flag. Uncertain use produces "Possible medication use—please confirm"; an ambiguous mapping requests clarification.
- Unavailable or denied medication sources produce "Medication history incomplete", including when other sources yield valid flags. A failed rule evaluation is visibly distinguished from no match.
- With the same medication and specialty, different procedure contexts follow the configured rules. An unknown procedure does not trigger an unsupported procedure-specific conclusion.
- Reviewed and Request clarification record the clinician, timestamp, flag version, and request. A materially changed flag needs renewed review, while the previous approved package and acknowledgment remain traceable.
- Repeated evaluation does not duplicate flags. No-match messages state coverage limits, and no flag recommends a medication or treatment change.

OCR accuracy, extraction accuracy, unsupported summary statements, relevant omissions, and clinician correction time should also be measured on a held-out sample. Numerical quality thresholds should be fixed after a representative baseline evaluation.

12. Decisions to Finalize Before Implementation

1. The single specialty, its clinician-reviewed referral checklist, supported medication classes/procedures, verified mapping source, rule reviewer, and initial approved rule set.
2. Supported document types, language, page limits, and upload limits.
3. Patient-link verification evidence and authorized verifier roles.
4. The chosen OCR/HTR integration and language model.
5. The exact FHIR version and supported fields.
6. Which missing items block submission and which allow a documented clinician override.
7. Retention periods and permitted central storage.
8. Evaluation dataset, available hardware, and measurable quality thresholds.

These decisions refine the included FYP capabilities without expanding the project into the larger roadmap.